

Rigid impression materials

By

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II- Rigid impression materials

1- Impression Compound

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Presentation:

- Sheets to make impression
- Sticks for border molding



<https://www.juniordentist.com/impression-compound.html>

• **Setting reaction:**

Hard $\xrightleftharpoons[37\text{ C}]{45-55\text{ C}}$ soft

- Inserted in mouth while it is warm at 45-55 C
- Then allowed to cool to mouth temperature at 37 C
- Thermoplastic material
- Reversible reaction

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- Type I is a lower-fusing material for border molding of an acrylic custom tray.
 - A compound stick may be softened over a flame, but the sheet should be immersed in warm water to ensure uniform softening of the material.
 - The compound should not be allowed to boil or ignite over the flame
 - Prolonged immersion or overheating in the water can alter the chemistry of the compound and the properties.
 - Cold water can be sprayed on the compound to assist in cooling.





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<http://removpros.dentistry.dal.ca>

Type II is a higher-fusing material in sheet form, also called tray compound, and can be used to make primary impressions as a tray to support a thin layer of a second impression material, such as ZOE wash paste, hydrocolloid, impression plaster, or elastomer, to make the secondary impression of the soft tissue. (wash technique)



https://www.researchgate.net/figure/Primary-impression-in-impression-compound_fig3_272370805

Uses:1-Used for making primary impression of edentulous patient by non perforated stock tray
2- border molding of special tray before taking final impression
3- impression by stock tray can be used for wash impression by Zn oxide /eugenol

Border molding



2- Zinc Oxide/eugenol

- **Presentation:**

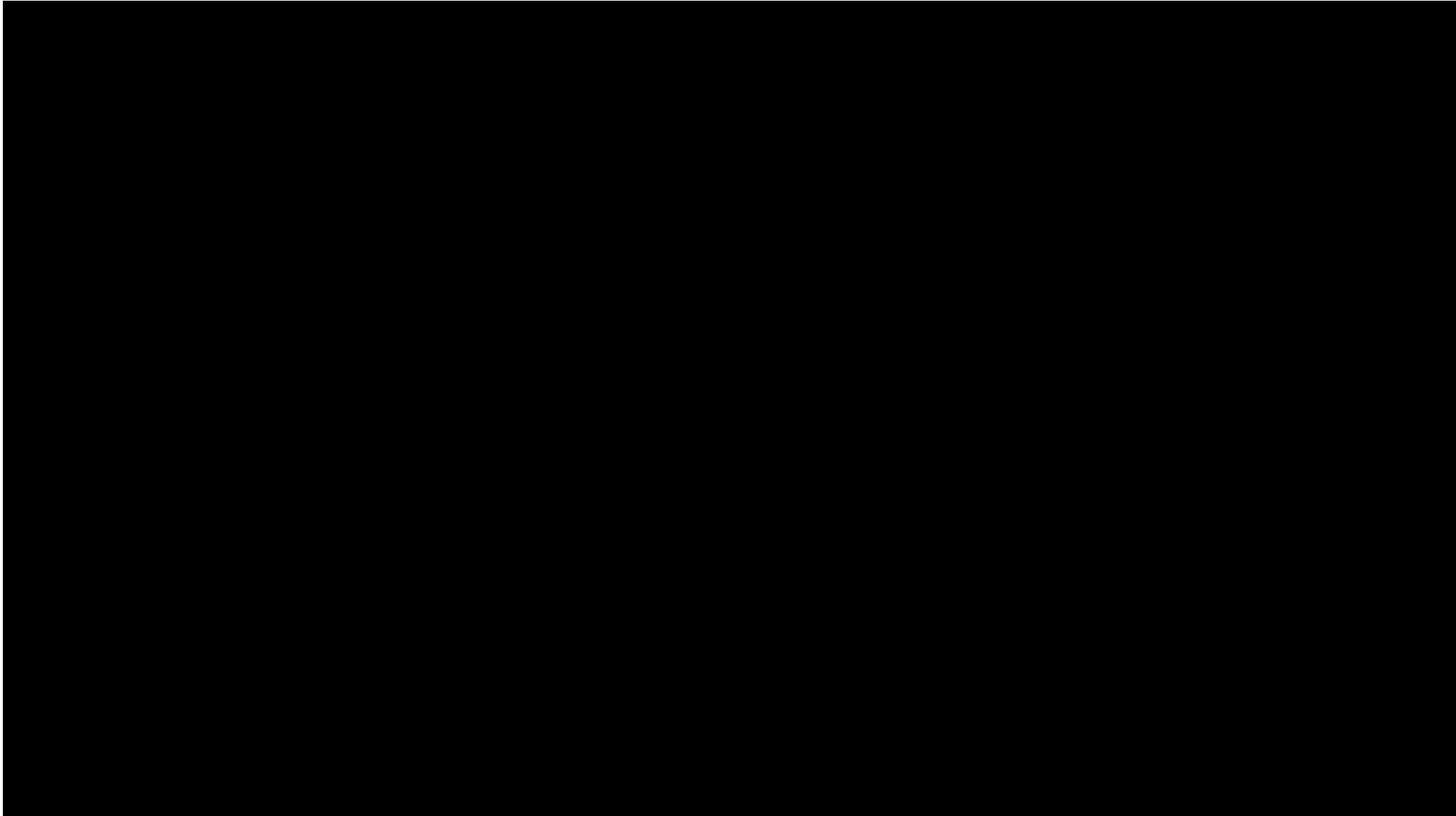
- Two collapsible metallic tubes (base, catalyst) with different colors
- Mixing on glass slab or oil resistant paper

ZOE impression pastes are dispensed as two pastes. **One tube** contains zinc oxide and vegetable or mineral oil; **the other** contains eugenol, rosin, filler.

the first reaction is the hydrolysis of zinc oxide to its hydroxide.
In formulations that do not contain water in the paste, the setting of the material is retarded until the mixed paste absorbs moisture in the patient's mouth.

These materials are classified as a hard paste (type I) or soft paste (type II)

Excellent dimensional accuracy and dimensional stability



Chelation

Setting reaction:



Setting reaction accelerators :

- Heat
- Humidity

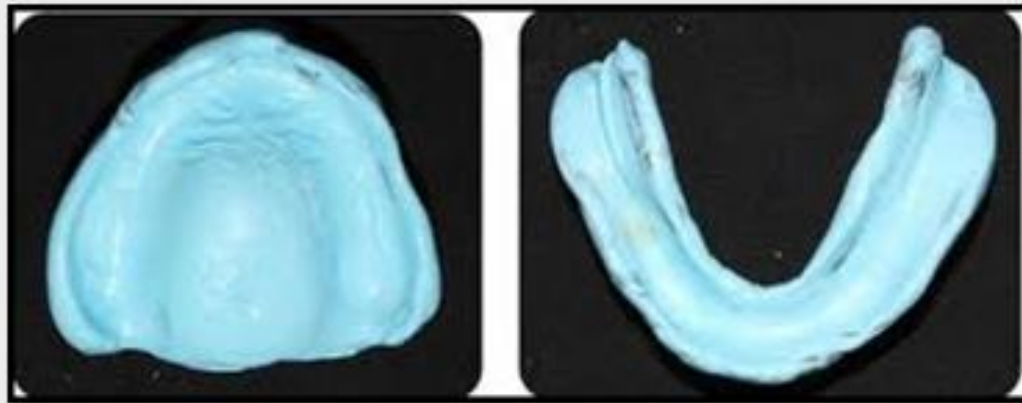
Eugenol causes:

- irritation to oral soft tissues**
- burning sensation**
- unpleasant taste to some patients**

-Also, Impression can adhere to face, lips of patient (used medical separator on skin before taking impression)

Uses

Used for making secondary impression of edentulous patient using special tray



Disinfection of Zinc oxide eugenol:

It is **contraindicated** to disinfect Zinc Oxide eugenol impression by sodium hypochlorite

It can be disinfected by iodophores

References:

- 1- Philips 13th edition
properties of dental
materials
- 2- Textbook of Applied dental
biomaterials- Cairo university